

The Coefficient Algebra of Exponentials of Rational Functions

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Abstract

Let R be a rational function with a pole of order m at a and principal part $\Pi(w) = \sum_{j=1}^m c_j w^{-j}$, $w = z - a$. The Laurent coefficients of e^R on a punctured disk about a are governed by the coefficients Λ_N of $\exp(\Pi)$, which satisfy the first-order recurrence $N \Lambda_N = \sum_{j=1}^{\min(m, N)} j c_j \Lambda_{N-j}$. We take this recurrence as the primitive object, construct from it a graded commutative Hopf algebra $\mathfrak{B}_m = \mathbb{C}[c_1, \dots, c_m]$ carrying the sequence (Λ_N) , and prove its structure theory: $\mathfrak{B}_m$ is canonically the quotient of the Hopf algebra of symmetric functions by the Hopf ideal $(p_{m+1}, p_{m+2}, \dots)$, under which Λ_N corresponds to the complete homogeneous symmetric function h_N , c_j to p_j/j , and the universal recurrence to Newton's identity. The sequence (Λ_N) is a divided-power sequence; as functions of c_1 the $N! \Lambda_N$ form an Appell sequence; the coefficient matrix is an element of the Appell subgroup of the exponential Riordan group; $N! \Lambda_N$ enumerates set partitions with block sizes at most m ; the graded Hopf automorphism group is the torus $(\mathbb{C}^\times)^m$; and the Hankel determinants of (Λ_N) are, up to explicit sign, rectangular Schur functions. We then prove that the analytic Laurent-coefficient construction is a model of the algebra, with quantitative growth bounds; we identify the classical specializations exactly (modified Bessel functions for $m=1$, Hermite-type polynomials for $m=2$, Wright generalized Bessel functions for single-term principal parts against a linear entire exponent) and refrain from claiming reductions that do not exist for $m \geq 3$; we prove holonomy of the full Laurent coefficient sequence at every pole, with an explicit recurrence and order bound derived from $y' = R' y$; and we prove an initiality theorem and a recognition theorem that together constitute the axiomatic characterization, with independence of the hypotheses established by counterexamples. The final section applies the algebra to the rough Heston characteristic function $\varphi_M = \exp(t A_M(x)/B_M(x))$, $x = t^\alpha$, produced by the diagonal Padé pipeline of the companion papers: we give the exact local Laurent data of φ_M at the zeros of B_M in the coefficient algebra, and prove convergence of Padé prices to exact prices under explicit hypotheses on a fixed Lewis-type contour, with no contour deformation through approximant poles. Statements that remain open (resurgent/alien-derivative structure, uniform-in-parameter domination, exponential-motive positioning) are listed as open problems, not theorems.

1 The universal recurrence

Throughout, $m \geq 1$ is a fixed integer and $c_1, \dots, c_m$ are independent commuting indeterminates over $\mathbb{C}$. Set

$$\mathfrak{B}_m := \mathbb{C}[c_1, \dots, c_m],$$

graded by $\text{wt}(c_j) = j$, so that $\mathfrak{B}_m = \bigoplus_{N \geq 0} (\mathfrak{B}_m)_N$ with $(\mathfrak{B}_m)_0 = \mathbb{C}$. All constructions in Sections 1–8 are exact identities in $\mathfrak{B}_m$ or in the formal power series ring $\mathfrak{B}_m[[v]]$; no analysis enters until Section 5.

Definition 1. $E(v) := \exp(\sum_{j=1}^m c_j v^j) \in \mathfrak{B}_m[[v]]$, and $\Lambda_N \in \mathfrak{B}_m$ is defined by $E(v) = \sum_{N \geq 0} \Lambda_N v^N$. We set $\Lambda_N := 0$ for $N < 0$.

The exponential is well defined because $\sum_j c_j v^j$ has zero constant term. Under the substitution $v = w^{-1}$, E becomes the formal expansion of $\exp(\sum_{j=1}^m c_j w^{-j})$, the exponential of the principal part of a rational function at a pole of order $\leq m$; the analytic counterpart is Theorem 23.

Lemma 2. [Universal recurrence] $\Lambda_0 = 1$ and, for all $N \geq 1$,

$$N \Lambda_N = \sum_{j=1}^{\min(m, N)} j c_j \Lambda_{N-j}. \quad (1)$$

Proof. E satisfies the linear first-order equation $E'(v) = (\sum_{j=1}^m j c_j v^{j-1}) E(v)$ in $\mathfrak{B}_m[[v]]$, obtained by formal differentiation of the exponential. The coefficient of v^{N-1} on the left is $N \Lambda_N$; on the right it is $\sum_{j=1}^m j c_j \Lambda_{N-j}$, in which terms with $j > N$ vanish by the convention $\Lambda_{<0} = 0$. The initial value $\Lambda_0 = E(0) = 1$. $\square$

Lemma 3. [Partition formula] For all $N \geq 0$,

$$\Lambda_N = \sum_{\substack{k_1, \dots, k_m \geq 0 \\ \sum_j j k_j = N}} \prod_{j=1}^m \frac{c_j^{k_j}}{k_j!}. \quad (2)$$

In particular Λ_N is homogeneous of weight N .

Proof. $E(v) = \prod_{j=1}^m \exp(c_j v^j) = \prod_{j=1}^m \sum_{k_j \geq 0} \frac{c_j^{k_j}}{k_j!} v^{j k_j}$; the product of the m series is computed by collecting the coefficient of v^N , which is (2). Homogeneity is immediate since the monomial $\prod c_j^{k_j}$ has weight $\sum j k_j = N$. $\square$

Proposition 4. [Bell-polynomial form] $N! \Lambda_N = B_N(1! c_1, 2! c_2, \dots, m! c_m, 0, \dots, 0)$, where B_N is the complete Bell polynomial, defined by $\exp(\sum_{j \geq 1} x_j t^j / j!) = \sum_{N \geq 0} B_N(x_1, \dots, x_N) t^N / N!$ (Comtet [Comtet, §3.3]).

Proof. Substitute $x_j = j! c_j$ for $j \leq m$ and $x_j = 0$ for $j > m$ in the defining generating identity of B_N ; the left side becomes $E(t)$. $\square$

Lemma 5. *[Derivations] For $1 \leq j \leq m$ and all N ,*

$$\frac{\partial \Lambda_N}{\partial c_j} = \Lambda_{N-j}.$$

Proof. $\partial_{c_j} E(v) = v^j E(v)$; compare coefficients of v^N . $\square$

Remark 6. Lemma 2 is exactly the Euler homogeneity relation: the grading operator $\mathcal{E} = \sum_{j=1}^m j c_j \partial_{c_j}$ satisfies $\mathcal{E} \Lambda_N = \sum_j j c_j \Lambda_{N-j}$ by Lemma 5, and $\mathcal{E} \Lambda_N = N \Lambda_N$ by Lemma 3. Conversely, any sequence of weight-homogeneous elements satisfying Lemma 5 with $\Lambda_0 = 1$ satisfies (1). The universal recurrence, the lowering operators, and the grading are one structure viewed three ways.

Theorem 7. *[Generation and freeness] $\mathbb{C}[\Lambda_1, \dots, \Lambda_m] = \mathfrak{B}_m$, and $\Lambda_1, \dots, \Lambda_m$ are algebraically independent over $\mathbb{C}$. Consequently $(\Lambda_N)_{N \geq 0} \subset \mathbb{C}[\Lambda_1, \dots, \Lambda_m]$: the entire coefficient sequence lives in the polynomial algebra on its first m terms.*

Proof. For $1 \leq N \leq m$, the multi-index $k_N = 1$, $k_j = 0$ ($j \neq N$) is the unique solution of $\sum j k_j = N$ using the variable c_N ; every other solution has $k_N = 0$ and involves only $c_1, \dots, c_{N-1}$. Hence $\Lambda_N = c_N + q_N(c_1, \dots, c_{N-1})$ with q_N polynomial. By induction, $c_N \in \mathbb{C}[\Lambda_1, \dots, \Lambda_N]$, so $\mathbb{C}[\Lambda_1, \dots, \Lambda_m] \supseteq \mathbb{C}[c_1, \dots, c_m]$; the reverse inclusion is trivial. Since $\text{trdeg}_{\mathbb{C}} \mathfrak{B}_m = m$ and $\Lambda_1, \dots, \Lambda_m$ generate it, they are algebraically independent. For $N > m$, $\Lambda_N \in \mathfrak{B}_m = \mathbb{C}[\Lambda_1, \dots, \Lambda_m]$; explicitly, iterate (1). $\square$

2 Hopf structure and the symmetric-function realization

Definition 8. Equip $\mathfrak{B}_m$ with the unique bialgebra structure for which each c_j is primitive: $\Delta c_j = c_j \otimes 1 + 1 \otimes c_j$, $\varepsilon(c_j) = 0$, extended as algebra morphisms. Since $\mathfrak{B}_m$ is graded and connected ($(\mathfrak{B}_m)_0 = \mathbb{C}$), it is automatically a Hopf algebra, with antipode determined by $S(c_j) = -c_j$ (graded connected bialgebras possess antipodes; see [MilnorMoore, §1]).

Theorem 9. *[Divided-power property] $\Delta \Lambda_N = \sum_{i=0}^N \Lambda_i \otimes \Lambda_{N-i}$ for all $N \geq 0$, $\varepsilon(\Lambda_N) = \delta_{N,0}$, and $\sum_{i=0}^N \Lambda_i S(\Lambda_{N-i}) = \delta_{N,0}$, with $S(\Lambda_N) = \Lambda_N(-c_1, \dots, -c_m)$.*

Proof. Apply the algebra morphism Δ coefficient-wise in $(\mathfrak{B}_m \otimes \mathfrak{B}_m)[[v]]$:

$$\sum_N \Delta \Lambda_N v^N = \Delta E(v) = \exp \left(\sum_j (c_j \otimes 1 + 1 \otimes c_j) v^j \right) = \exp \left(\sum_j (c_j \otimes 1) v^j \right) \exp \left(\sum_j (1 \otimes c_j) v^j \right),$$

the factorization being valid because the two exponents commute. The right side equals $(\sum_i \Lambda_i \otimes 1 v^i)(\sum_k 1 \otimes \Lambda_k v^k)$; compare coefficients. The counit and antipode identities follow the same way from $\varepsilon E(v) = 1$ and $SE(v) = E(v)^{-1} = \exp(-\sum c_j v^j)$. $\square$

Thus (Λ_N) is a *divided-power sequence* in the graded connected Hopf algebra $\mathfrak{B}_m$: $\Lambda_N \in (\mathfrak{B}_m)_N$, $\Lambda_0 = 1$, $\Delta \Lambda_N = \sum \Lambda_i \otimes \Lambda_{N-i}$. Equivalently, the series $E(v)$ is group-like in the completed Hopf algebra $\mathfrak{B}_m[[v]]$.

Theorem 10. *[Primitives] $\text{Prim}(\mathfrak{B}_m) = \bigoplus_{j=1}^m \mathbb{C} c_j$; in particular the primitive space in weight j is one-dimensional for $1 \leq j \leq m$ and zero for $j > m$.*

Proof. Each c_j is primitive by construction. Conversely, let $f = \sum_{\alpha} a_{\alpha} c^{\alpha}$ be primitive, with multi-index notation $c^{\alpha} = \prod_j c_j^{\alpha_j}$. Then $\Delta(c^{\alpha}) = \prod_j (c_j \otimes 1 + 1 \otimes c_j)^{\alpha_j} = \sum_{0 \leq \beta \leq \alpha} \binom{\alpha}{\beta} c^{\beta} \otimes c^{\alpha-\beta}$. The component of $\Delta f - f \otimes 1 - 1 \otimes f$ in $\mathfrak{B}_m^+ \otimes \mathfrak{B}_m^+$ ($\mathfrak{B}_m^+ := \ker \varepsilon$) is $\sum_{\alpha} a_{\alpha} \sum_{0 < \beta < \alpha} \binom{\alpha}{\beta} c^{\beta} \otimes c^{\alpha-\beta}$. The elements $\{c^{\beta} \otimes c^{\gamma} : \beta, \gamma \neq 0\}$ are linearly independent in $\mathfrak{B}_m \otimes \mathfrak{B}_m$, and for fixed (β, γ) the total coefficient is $a_{\beta+\gamma} \binom{\beta+\gamma}{\beta}$. Primitivity therefore forces $a_{\alpha} = 0$ whenever α admits a decomposition $\alpha = \beta + \gamma$ with $\beta, \gamma \neq 0$, i.e. whenever $|\alpha| := \sum_j \alpha_j \geq 2$. Hence f is a linear combination of $1, c_1, \dots, c_m$, and $\varepsilon(f) = 0$ kills the constant. $\square$

2.1 The symmetric-function isomorphism

Let $\text{Sym} = \mathbb{C}[p_1, p_2, \dots]$ be the Hopf algebra of symmetric functions over $\mathbb{C}$, with power sums p_r primitive, complete homogeneous symmetric functions h_N , and Schur functions s_{λ} (Macdonald [Macdonald, Ch. I]). Write $\mathfrak{B}_{\infty} := \mathbb{C}[c_1, c_2, \dots]$ for the evident $m = \infty$ version of Definitions 1 and 8; Lemmas 2–5 and Theorems 7–10 hold verbatim with $\min(m, N)$ replaced by N .

Theorem 11. *[Symmetric-function realization] The assignment $\Psi(p_j) := j c_j$ extends uniquely to an isomorphism of graded Hopf algebras $\Psi: \text{Sym} \xrightarrow{\sim} \mathfrak{B}_{\infty}$, and*

$$\Psi(h_N) = \Lambda_N \quad (N \geq 0).$$

Under Ψ , Newton's identity $N h_N = \sum_{j=1}^N p_j h_{N-j}$ becomes the universal recurrence (1). Moreover the ideal $I_m := (c_{m+1}, c_{m+2}, \dots) \subset \mathfrak{B}_{\infty}$ is a graded Hopf ideal, and

$$\mathfrak{B}_m \cong \mathfrak{B}_{\infty} / I_m \cong \text{Sym} / (p_{m+1}, p_{m+2}, \dots)$$

as graded Hopf algebras, compatibly with the sequences: $h_N \mapsto \Lambda_N$ throughout.

Proof. Sym and $\mathfrak{B}_\infty$ are free commutative algebras on primitive generators p_j , resp. c_j , one in each weight $j \geq 1$; Ψ matches them by weight with the nonzero scalar j , hence is an isomorphism of graded algebras. It is a coalgebra map because it sends primitives to primitives and both coproducts are algebra morphisms determined on generators; a bialgebra isomorphism of Hopf algebras automatically respects antipodes. For the sequence: the generating identity $\sum_{N \geq 0} h_N t^N = \exp(\sum_{r \geq 1} p_r t^r / r)$ ([Macdonald, I.(2.10)]) maps under Ψ (applied coefficient-wise in t) to $\sum_N \Psi(h_N) t^N = \exp(\sum_r c_r t^r) = \sum_N \Lambda_N t^N$, so $\Psi(h_N) = \Lambda_N$. Newton's identity is [Macdonald, I.(2.11)]; applying Ψ gives (1) (and, independently, Lemma 2 re-proves Newton's identity via the differential equation $E' = (\sum j c_j v^{j-1}) E$). For the quotient: I_m is generated by primitives, so $\Delta I_m \subseteq I_m \otimes \mathfrak{B}_\infty + \mathfrak{B}_\infty \otimes I_m$, $\varepsilon(I_m) = 0$, $S(I_m) \subseteq I_m$: a Hopf ideal. The quotient map $\mathfrak{B}_\infty \rightarrow \mathfrak{B}_m$, $c_j \mapsto c_j$ ($j \leq m$), $c_j \mapsto 0$ ($j > m$), carries $\Lambda_N^{(\infty)}$ to Λ_N by Lemma 3 (multi-indices with $k_j > 0$ for some $j > m$ die). $\square$

Remark 12. $\mathfrak{B}_\infty$ is *not* the Faà di Bruno Hopf algebra: the latter is the coordinate Hopf algebra of the group of formal diffeomorphisms and is not cocommutative (Figueroa-Gracia-Bondía [FGB]), whereas $\mathfrak{B}_\infty \cong \text{Sym}$ is commutative and cocommutative. The relation to Bell polynomials is the coefficient identity of Proposition 4, not an isomorphism with the Faà di Bruno object. This answers, precisely, the question “is it the Bell algebra”: the coefficient algebra is the algebra of symmetric functions, in which the Λ_N are the complete homogeneous basis re-expressed through rescaled power sums.

Remark 13. [Hall algebra] The classical Hall algebra of finite modules over a discrete valuation ring with finite residue field is isomorphic, after the standard rescaling, to Sym ([Macdonald, Ch. II–III]). Via Theorem 11, $\mathfrak{B}_\infty$ therefore *does* arise as a Hall algebra, and $\mathfrak{B}_m$ as an explicit Hopf quotient of one. We record this as an identification available by citation; we make no use of the finer (Hall–Littlewood, q -) structure here, and lifting the coefficient sequence (Λ_N) to the q -deformed level is Open Problem 4.

3 Combinatorial and matrix realizations

Theorem 14. [Species / set-partition interpretation] For $N \geq 1$,

$$N! \Lambda_N = \sum_{\substack{\pi \vdash [N] \\ \text{all blocks of size} \leq m}} \prod_{B \in \pi} |B|! c_{|B|},$$

the sum over set partitions π of $\{1, \dots, N\}$ with block sizes at most m . Equivalently, $E(v)$ is the exponential generating function of the species “sets of blocks,” each block of size $j \leq m$ weighted $j! c_j$.

Proof. The number of set partitions of $[N]$ with exactly k_j blocks of size j is $N! / \prod_j ((j!)^{k_j} k_j!)$. Hence

$$\sum_{\pi} \prod_{B \in \pi} |B|! c_{|B|} = \sum_{k: \sum j k_j = N} \frac{N!}{\prod_j (j!)^{k_j} k_j!} \prod_j (j! c_j)^{k_j} = N! \sum_{k: \sum j k_j = N} \prod_j \frac{c_j^{k_j}}{k_j!} = N! \Lambda_N$$

by Lemma 3. The species statement is the exponential formula (Flajolet–Sedgewick [FS, II.2]; Bergeron–Labelle–Leroux [BLL]). $\square$

Theorem 15. [Appell structure] Fix $c_2, \dots, c_m$ and regard $p_N(c_1) := N! \Lambda_N$ as a polynomial in c_1 . Then $(p_N)_{N \geq 0}$ is an Appell sequence: $p'_N(c_1) = N p_{N-1}(c_1)$, with exponential generating function $\sum_N p_N(c_1) t^N / N! = e^{c_1 t} A(t)$, $A(t) = \exp(\sum_{j=2}^m c_j t^j)$.

Proof. Lemma 5 with $j = 1$ gives $\partial_{c_1} \Lambda_N = \Lambda_{N-1}$, i.e. $p'_N = N p_{N-1}$; the generating function is Definition 1 with the c_1 -factor split off. Appell sequences are exactly the Sheffer sequences with identity delta series, so (p_N) is Sheffer of Appell type (Roman [Roman, §2.5]). $\square$

Theorem 16. [Riordan-group realization] The lower-triangular matrix $D = (d_{n,k})_{n,k \geq 0}$, $d_{n,k} := \frac{n!}{k!} \Lambda_{n-k}$, is the exponential Riordan array $(E(t), t)$, an element of the Appell subgroup of the exponential Riordan group. The group law restricted to this subgroup, $(E(t), t) \cdot (E'(t), t) = (E(t) E'(t), t)$, corresponds to addition $c_j \mapsto c_j + c'_j$ of parameters, i.e. to the coproduct of Theorem 9.

Proof. By definition (Shapiro et al. [Shapiro]; Barry [Barry]) the exponential Riordan array $(g(t), f(t))$ has entries $d_{n,k} = \frac{n!}{k!} [t^{n-k}] g(t) f(t)^k / t^k$ reduced, for $f(t) = t$, to $d_{n,k} = \frac{n!}{k!} [t^{n-k}] g(t)$; with $g = E$ this is $\frac{n!}{k!} \Lambda_{n-k}$. Matrix multiplication of Appell arrays multiplies the g 's; $E(t; c) E(t; c') = E(t; c + c')$ since the exponents add. The identification with the coproduct is Theorem 9 read through $\Lambda_N(c + c') = \sum_i \Lambda_i(c) \Lambda_{N-i}(c')$, which is the image of $\Delta \Lambda_N$ under evaluation at (c, c') . $\square$

3.1 Hankel determinants are rectangular Schur functions

Theorem 17. [Hankel–Schur identity] In $\mathfrak{B}_\infty \cong \text{Sym}$ (hence, after applying the quotient of Theorem 11, as identities in $\mathfrak{B}_m$), for every $n \geq 1$:

$$\det (\Lambda_{i+j})_{0 \leq i, j \leq n-1} = (-1)^{\binom{n}{2}} \Psi(s_{((n-1)^n)}), \quad \det (\Lambda_{i+j+1})_{0 \leq i, j \leq n-1} = (-1)^{\binom{n}{2}} \Psi(s_{(n^n)}),$$

where (r^n) denotes the $n \times r$ rectangular partition.

Proof. Work in Sym with $\Lambda_N = h_N$ (Theorem 11). Reverse the rows of the Hankel matrix by $i \mapsto n-1-i$; this multiplies the determinant by the sign of the reversal permutation, $(-1)^{\binom{n}{2}}$. In 1-based indices (i, j) the reversed matrix has entries $h_{n-1-i+j}$ (first case), which is the Jacobi–Trudi matrix $(h_{\lambda_i - i + j})$ for the constant partition $\lambda_i = n-1$; by the Jacobi–Trudi identity ([Macdonald, I.(3.4)]) its determinant is $s_{((n-1)^n)}$. The shifted case is identical with $\lambda_i = n$. $\square$

Corollary 18. *[Normality of the Padé table of E] Let $\text{ev}_c: \mathfrak{B}_m \rightarrow \mathbb{C}$ be evaluation at $c \in \mathbb{C}^m$ and let $\mathcal{L}$ be the moment functional $\mathcal{L}[x^N] = \text{ev}_c(\Lambda_N)$. The Hankel determinants governing the diagonal Padé approximants of $E(v)$ at c (Gragg [Gragg]) are the numbers $(-1)^{\binom{n}{2}} s_{((n-1)^n)}$ and $(-1)^{\binom{n}{2}} s_{(n^n)}$ evaluated under $p_j = j c_j$ ($j \leq m$), $p_j = 0$ ($j > m$). In particular, for $m = 1$ (so $p_1 = c_1$, $p_{\geq 2} = 0$), the exponential specialization $s_\lambda \mapsto f^\lambda c_1^{|\lambda|} / |\lambda|!$ ([Macdonald, I. §7, Ex. 24(a) with $n \rightarrow \infty$; equivalently ev of [Macdonald, I.(5.16)] at the exponential specialization]), where $f^\lambda > 0$ is the number of standard Young tableaux, shows every such determinant is nonzero for $c_1 \neq 0$: the Padé table of $e^{c_1 v}$ is normal, recovering the classical normality of the exponential.*

Proof. Immediate from Theorem 17, the Padé–Hankel dictionary [Gragg], and $f^\lambda \geq 1$ for every partition λ . $\square$

Remark 19. For $m \geq 2$ the quasi-definiteness locus of $\mathcal{L}$ is precisely the complement of the union of the hypersurfaces $\{c \in \mathbb{C}^m: s_{((n-1)^n)}(p_j = j c_j) = 0\}$; the obstruction to Padé normality of exponentials of principal parts is a rectangular Schur divisor. This gives the “discriminant locus” appearing in the companion Chebyshev–Wheeler paper a name in the pure-pole case.

4 Automorphisms and differential Galois structure

Theorem 20. *[Graded Hopf automorphisms] $\text{Aut}_{\text{gr-Hopf}}(\mathfrak{B}_m) \cong (\mathbb{C}^\times)^m$, acting by $c_j \mapsto \lambda_j c_j$. The subgroup induced by the geometric rescalings $w \mapsto \lambda w$ of the local coordinate is the one-parameter diagonal $\lambda_j = \lambda^j$.*

Proof. A graded Hopf automorphism preserves $\text{Prim}(\mathfrak{B}_m)$ and each of its graded pieces, which are the lines $\mathbb{C}c_j$ by Theorem 10; hence $c_j \mapsto \lambda_j c_j$ with $\lambda_j \in \mathbb{C}^\times$, and any such assignment plainly defines a graded Hopf automorphism. Under $w \mapsto \lambda w$ the principal part $\sum c_j w^{-j}$ becomes $\sum (c_j \lambda^{-j}) w^{-j}$, giving the stated one-parameter subgroup (with λ^{-j} ; reparametrize). $\square$

Proposition 21. *[Picard–Vessiot group] Let $R \in \mathbb{C}(z)$ be nonconstant. Then $y = e^R$ is transcendental over $\mathbb{C}(z)$ and the Picard–Vessiot group of $y' = R' y$ over $\mathbb{C}(z)$ is $\mathbb{G}_m$. The group acts on solutions by $y \mapsto \lambda y$, hence on the full Laurent coefficient sequence (A_n) of Section 5 diagonally by $A_n \mapsto \lambda A_n$: the coefficient sequence spans the weight-one isotypic line, while the algebra $\mathfrak{B}_m$ itself is Galois-invariant.*

Proof. The PV group of a rank-one equation is a Zariski-closed subgroup of $\mathbb{G}_m$, hence $\mathbb{G}_m$ itself or the finite group μ_n ; the latter occurs iff $y^n \in \mathbb{C}(z)$ for some $n \geq 1$ (*van der Put–Singer* [vdPS, Ex. 1.19, §1.4]). Suppose $e^{nR} = f \in \mathbb{C}(z)$. Then $nR' = f'/f$. Every residue of R' vanishes, since R' is the derivative of a rational function; every residue of f'/f at a zero or pole of f is the (nonzero) integer order of f there. Hence f has no zeros or poles, so f is constant and $R' = 0$, contradicting R nonconstant. $\square$

5 The analytic model

We now verify that the formal object of Sections 1–3 computes actual Laurent coefficients, with quantitative control. Fix $c = (c_1, \dots, c_m) \in \mathbb{C}^m$ and write $\Lambda_N(c) := \text{ev}_c(\Lambda_N)$, $C := \sum_{j=1}^m |c_j|$.

Proposition 22. *[Growth bound] For all $N \geq 0$ and all $v > 0$, $|\Lambda_N(c)| \leq \Lambda_N(|c_1|, \dots, |c_m|) \leq v^{-N} \exp(\sum_j |c_j| v^j)$. Consequently, for $N \geq mC$,*

$$|\Lambda_N(c)| \leq e^{N/m} \left(\frac{mC}{N} \right)^{N/m},$$

so $|\Lambda_N(c)|^{1/N} \rightarrow 0$ super-geometrically, at the rate of an entire function of order m in $1/w$.

Proof. The first inequality is termwise from (2). For the second, all terms of $\sum_M \Lambda_M(|c|) v^M = \exp(\sum_j |c_j| v^j)$ are nonnegative, so any single term is bounded by the sum: $\Lambda_N(|c|) v^N \leq \exp(\sum_j |c_j| v^j) \leq e^{Cv^m}$ for $v \geq 1$. Choose $v = (N/(mC))^{1/m} \geq 1$; then $e^{Cv^m} = e^{N/m}$ and $v^{-N} = (mC/N)^{N/m}$. $\square$

Theorem 23. *[Pure-pole expansion] For every $c \in \mathbb{C}^m$ and $w \neq 0$, the series $\sum_{N \geq 0} \Lambda_N(c) w^{-N}$ converges absolutely, uniformly on $|w| \geq \epsilon$ for each $\epsilon > 0$, and*

$$\exp\left(\sum_{j=1}^m c_j w^{-j}\right) = \sum_{N \geq 0} \Lambda_N(c) w^{-N}.$$

In particular $\Lambda_N(c)$ is the coefficient of w^{-N} in the Laurent expansion of e^Π on $\mathbb{C}^\times$, and the map $c \mapsto (\Lambda_N(c))_{N \geq 0}$ is injective.

Proof. Expanding each factor $\exp(c_j w^{-j})$ and multiplying, the resulting multi-series is dominated termwise by $\prod_j \sum_{k_j} |c_j|^{k_j} |w|^{-j k_j} / k_j! = \exp(\sum_j |c_j| |w|^{-j}) < \infty$, so it converges absolutely and may be rearranged by total degree, yielding $\sum_N \Lambda_N(c) w^{-N}$ with the coef-

ficients of Lemma 3. Uniqueness of Laurent coefficients on an annulus identifies them with the Laurent coefficients of e^Π . Injectivity: by the triangularity in the proof of Theorem 7, c_N is a polynomial in $\Lambda_1(c), \dots, \Lambda_N(c)$. $\square$

Theorem 24. [Local Laurent structure of e^R ; the coefficient module] Let $R \in \mathbb{C}(z)$ have a pole at a of order $m \geq 1$ with principal part $\Pi(w) = \sum_{j=1}^m c_j w^{-j}$, $w = z - a$, $c_m \neq 0$, and let $r_a \in (0, \infty]$ be the distance from a to the nearest other pole of R . Write $H := R - \Pi$, analytic on $|w| < r_a$, and $e^{H(a+w)} = \sum_{l \geq 0} b_l w^l$, $\limsup_l |b_l|^{1/l} \leq 1/r_a$. Then on $0 < |w| < r_a$,

$$e^{R(a+w)} = \sum_{n \in \mathbb{Z}} A_n w^n, \quad A_n = \sum_{k \geq \max(0, -n)} \Lambda_k(c) b_{n+k},$$

each A_n given by an absolutely convergent series. The map $\mathfrak{B}_m \times \mathbb{C}[[w]]_{r_a} \rightarrow \mathbb{C}^{\mathbb{Z}}$ so defined is $\mathbb{C}$ -bilinear in the coefficient sequences $(\Lambda_k(c))$ and (b_l) : the Laurent coefficients of e^R constitute a module over the coefficient algebra, with structure constants supported on the exponent lattice $n = l - k$.

Proof. $e^R = e^\Pi e^H$ exactly, since Π and H are scalar functions. On any circle $|w| = \rho \in (0, r_a)$, both factors are given by absolutely convergent Laurent/Taylor series (Theorem 23; Cauchy), so their product's Laurent coefficients are the Cauchy convolutions $A_n = \sum_k \Lambda_k(c) b_{n+k}$ (with $b_{<0} := 0$), and the double series converges absolutely because, choosing $\rho' \in (\rho, r_a)$, $|b_{n+k}| \leq B \rho'^{-(n+k)}$ while $|\Lambda_k(c)|$ decays super-geometrically by Proposition 22: $\sum_k |\Lambda_k(c)| B \rho'^{-(n+k)} \leq B \rho'^{-n} \sum_k e^{k/m} (mC/k)^{k/m} \rho'^{-k} < \infty$. Bilinearity is manifest. $\square$

Proposition 25. [Truncation error] In the setting of Theorem 24, fix $\rho' < r_a$ with $|b_l| \leq B \rho'^{-l}$, and let $K \geq K_0 := \max\{mC, e mC (2/\rho')^m\}$. Then

$$\left| A_n - \sum_{k=\max(0, -n)}^K \Lambda_k(c) b_{n+k} \right| \leq B \rho'^{-n} 2^{-K}.$$

Proof. For $k \geq K_0$, Proposition 22 gives $|\Lambda_k(c)| \rho'^{-k} \leq [(e mC/k)^{1/m} / \rho']^k \leq 2^{-k}$ by the choice of K_0 . Sum the geometric tail $\sum_{k > K} 2^{-k} \leq 2^{-K}$ against $B \rho'^{-n}$ (absorbing $\rho'^{-k} \leq$ what was already used). $\square$

Proposition 26. [Hadamard closure] For $c, c' \in \mathbb{C}^m$, the Hadamard product sequence $(\Lambda_N(c) \Lambda_N(c'))_N$ is P -recursive (holonomic).

Proof. Each factor satisfies the polynomial-coefficient recurrence (1); the Hadamard product of P -recursive sequences is P -recursive (Stanley [StanleyDF]; Lipshitz [Lipshitz]). $\square$

6 Classical specializations

The identifications below are exact and exhaustive in the stated cases; no claim is made that a general principal part of order $m \geq 3$ reduces to a single classical one-variable function. The general case is the algebra itself.

Theorem 27. [$m = 1$: modified Bessel] For $b, c \in \mathbb{C}$, the Laurent expansion of $e^{c/w+bw}$ on $\mathbb{C}^\times$ has coefficients

$$[w^{-n}] e^{c/w+bw} = \sum_{j \geq 0} \frac{c^{j+n} b^j}{j! (j+n)!} \quad (n \geq 0), \quad [w^n] e^{c/w+bw} = \text{same with } b \leftrightarrow c.$$

For any determination $\sqrt{bc}$ of a square root of bc with $(\sqrt{bc})^2 = bc$, these equal $(c/b)^{n/2} I_n(2\sqrt{bc})$ (interpreting $(c/b)^{n/2} := c^n (bc)^{-n/2}$ with the same determination), where $I_n(x) = \sum_{j \geq 0} (x/2)^{2j+n} / (j! (j+n)!)$. Setting $b = c = x/2$, $w = 1/t$ recovers the classical generating identity $\exp\left(\frac{x}{2}(t + 1/t)\right) = \sum_{n \in \mathbb{Z}} I_n(x) t^n$.

Proof. Here $\Lambda_k(c) = c^k / k!$ and $b_l = b^l / l!$; Theorem 24 with $r_a = \infty$ gives $A_{-n} = \sum_{k \geq n} \frac{c^k}{k!} \frac{b^{k-n}}{(k-n)!}$; substitute $k = j + n$. The Bessel form is the same series regrouped: $\sum_j (bc)^j / (j! (j+n)!) = (bc)^{-n/2} I_n(2\sqrt{bc})$ term by term, both sides being unambiguous power series in bc once the prefactor convention is fixed. Symmetry $w \mapsto 1/w$ exchanges b, c and $n \mapsto -n$, giving the positive coefficients and, with $I_{-n} = I_n$, the classical identity (cf. [DLMF, 10.35.1]). $\square$

Theorem 28. [$m = 2$: Hermite-type polynomials] Define $\mathbf{H}_N(x, y)$ by $\exp(xt + yt^2) = \sum_{N \geq 0} \mathbf{H}_N(x, y) t^N / N!$ (the Hermite–Kampé de Fériet polynomials [Appell]). Then for $m = 2$,

$$\Lambda_N(c_1, c_2) = \frac{\mathbf{H}_N(c_1, c_2)}{N!} = \sum_{0 \leq k \leq N/2} \frac{c_1^{N-2k} c_2^k}{(N-2k)! k!},$$

and $\mathbf{H}_N(x, y) = (i\sqrt{y})^N H_N(x / (2i\sqrt{y}))$ for any fixed square root, where H_N is the classical (physicists') Hermite polynomial.

Proof. The first two equalities are Definition 1 for $m = 2$ and Lemma 3. The classical relation follows by comparing $\exp(xt + yt^2)$ with $\exp(2Xs - s^2) = \sum H_N(X) s^N / N!$ under $s = i\sqrt{y} t$, $X = x / (2i\sqrt{y})$; both sides are polynomial identities in $(x, y^{1/2})$, hence hold for every branch choice consistently. $\square$

Theorem 29. [Single-term pole of order m : Wright functions] Let $\phi(\rho, \beta; \zeta) := \sum_{k \geq 0} \frac{\zeta^k}{k! \Gamma(\rho k + \beta)}$ be Wright's generalized Bessel function (Wright [Wright1933]; Kilbas–Saigo [KilbasSaigo, §1.11]), entire in ζ for $\rho > -1$. Then for $b \in \mathbb{C}^\times$, $c \in \mathbb{C}$, $n \in \mathbb{Z}$,

$$[w^{-n}] \exp\left(\frac{c}{w^m} + bw\right) = b^{-n} \phi(m, 1-n; c b^m).$$

Proof. Here $\Lambda_{mk} = c^k / k!$ and $\Lambda_N = 0$ otherwise, $b_l = b^l / l!$ ($l \geq 0$), $b_l := 0$ ($l < 0$). Theorem 24: $A_{-n} = \sum_k \frac{c^k}{k!} \frac{b^{mk-n}}{(mk-n)!}$ over k with $mk \geq n$. Since $1/\Gamma(\ell) = 0$ for $\ell \in \mathbb{Z}_{\leq 0}$, the constraint is absorbed by writing $(mk-n)! = \Gamma(mk-n+1) = \Gamma(mk+(1-n))$, and the full sum over $k \geq 0$ equals $b^{-n} \sum_k (cb^m)^k / (k! \Gamma(mk+1-n))$. $\square$

Remark 30. [Scope of special-function reductions] Theorems 27–29 exhaust the cases in which a single classical family arises from a natural one-parameter entire factor: $m=1$, $m=2$, and pure top-order poles. For a general principal part with $m \geq 3$ and mixed c_j , the coefficient A_n is the genuine multivariate partition sum of Theorem 24, i.e. an element of the $\mathfrak{B}_m$ -module, and we do not assert any reduction to Wright, Fox–Wright, or Fox H functions of one variable. What replaces such a reduction is the structure theory of Sections 2–3 together with the holonomy of Section 7: closed recurrences, not closed one-variable forms.

7 Holonomy

Theorem 31. [*D-finiteness and the coefficient recurrence*] Let $R \in \mathbb{C}(z)$ be nonconstant, write $R' = U/V$ in lowest terms with $U, V \in \mathbb{C}[z]$, and let a be a pole of R of order $m \geq 1$. Set $\tilde{V}(w) := V(a+w) = \sum_{p=0}^{d_V} V_p w^p$, $\tilde{U}(w) := U(a+w) = \sum_{p=0}^{d_U} U_p w^p$. Then $y = e^R$ satisfies the first-order equation $Vy' - Uy = 0$ over $\mathbb{C}(z)$ (in particular y is D -finite), and the full two-sided Laurent coefficient sequence $(A_n)_{n \in \mathbb{Z}}$ of y at a (Theorem 24) satisfies, for every $s \in \mathbb{Z}$,

$$\sum_{p=0}^{d_V} V_p (s+1-p) A_{s+1-p} = \sum_{p=0}^{d_U} U_p A_{s-p}. \quad (3)$$

This is a linear recurrence with coefficients affine in s , of order at most $\max(d_V, d_U + 1)$, valid in both directions $n \rightarrow \pm\infty$. Moreover $w^{m+1} | \tilde{V}$ (indeed $\text{ord}_{w=0} R' = -(m+1)$ forces $\text{ord} \tilde{V} - \text{ord} \tilde{U} = m+1$), and in the pure-pole case $R = \Pi$ the recurrence (3) reduces exactly to the universal recurrence (1).

Proof. $y' = R' y$ is immediate; clearing denominators gives $Vy' = Uy$ with polynomial coefficients, which is the definition of D -finiteness for y (Stanley [StanleyDF]). On $0 < |w| < r_a$ insert $y = \sum_n A_n w^n$ (absolutely and locally uniformly convergent, so termwise differentiation and Cauchy products with the polynomials $\tilde{V}, \tilde{U}$ are valid): $\sum_p V_p w^p \sum_n n A_n w^{n-1} = \sum_p U_p w^p \sum_n A_n w^n$. Equate coefficients of w^s : $\sum_p V_p (s+1-p) A_{s+1-p} = \sum_p U_p A_{s-p}$, which is (3). The order statement counts the index span. The valuation claim: near a , $R' = -m c_m w^{-(m+1)} + O(w^{-m})$ with $c_m \neq 0$, so $\text{ord}_{w=0}(\tilde{U}/\tilde{V}) = -(m+1)$; since $\gcd(U, V) = 1$, at most one of $\tilde{U}(0), \tilde{V}(0)$ vanishes, forcing $\tilde{U}(0) \neq 0$ and $w^{m+1} \parallel \tilde{V}$. Pure pole: $R = \Pi$ gives $R' = -\sum_j j c_j w^{-j-1} = \tilde{U}/\tilde{V}$ with $\tilde{V} = w^{m+1}$, $\tilde{U} = -\sum_{j=1}^m j c_j w^{m-j}$; then (3) reads $(s-m) A_{s-m} = -\sum_j j c_j A_{s-m+j}$, and putting $A_{-N} = \Lambda_N$, $s-m = -N$: $N \Lambda_N = \sum_j j c_j \Lambda_{N-j}$. $\square$

Corollary 32. *For fixed $\mathbf{c}$, every coefficient sequence in the module of Theorem 24 is P -recursive with an explicitly computable recurrence (3); creative-telescoping toolchains apply directly (Kauers–Paule [KauersPaule]; Koutschan [Koutschan]).*

8 Axiomatic characterization

Following the constructive development, we now characterize what has been built. Two statements are proved: an *initiality* theorem in the category naturally attached to the universal recurrence, and a *recognition* theorem identifying $\mathfrak{B}_m$ among graded Hopf algebras. The hypotheses of the recognition theorem are shown to be independent.

8.1 Initiality

Definition 33. *[The category $\mathcal{C}_m$] Objects of $\mathcal{C}_m$ are triples $(A, \gamma = (\gamma_1, \dots, \gamma_m) \in A^m, (\lambda_N)_{N \geq 0})$ where A is a commutative unital $\mathbb{C}$ -algebra, $\lambda_0 = 1$, and*

$$N \lambda_N = \sum_{j=1}^{\min(m, N)} j \gamma_j \lambda_{N-j} \quad (N \geq 1). \quad (4)$$

A morphism $(A, \gamma, \lambda) \rightarrow (A', \gamma', \lambda')$ is a unital $\mathbb{C}$ -algebra homomorphism $\varphi: A \rightarrow A'$ with $\varphi(\gamma_j) = \gamma'_j$ for all j and $\varphi(\lambda_N) = \lambda'_N$ for all N .

Theorem 34. *[Initiality] $(\mathfrak{B}_m, \mathbf{c}, \Lambda)$ is an initial object of $\mathcal{C}_m$: for every object (A, γ, λ) there is exactly one morphism $\varphi: (\mathfrak{B}_m, \mathbf{c}, \Lambda) \rightarrow (A, \gamma, \lambda)$.*

Proof. *Existence.* Since $\mathfrak{B}_m$ is the free commutative $\mathbb{C}$ -algebra on $c_1, \dots, c_m$, there is a unique algebra map φ with $\varphi(c_j) = \gamma_j$. We claim $\varphi(\Lambda_N) = \lambda_N$ for all N , by strong induction: true at $N = 0$ ($\Lambda_0 = 1 \mapsto 1 = \lambda_0$); for $N \geq 1$, since N is invertible in the $\mathbb{C}$ -algebra A ,

$$\varphi(\Lambda_N) = \varphi\left(\frac{1}{N} \sum_j j c_j \Lambda_{N-j}\right) = \frac{1}{N} \sum_j j \gamma_j \varphi(\Lambda_{N-j}) = \frac{1}{N} \sum_j j \gamma_j \lambda_{N-j} = \lambda_N,$$

using Lemma 2 in $\mathfrak{B}_m$, the inductive hypothesis, and (4) in A . *Uniqueness.* Any morphism must send $c_j \mapsto \gamma_j$, and an algebra map out of a polynomial ring is determined by the images of the variables; the λ -compatibility is then automatic by the computation just made, so no further freedom exists. $\square$

Remark 35. Equation (4) determines (λ_N) uniquely from γ in any $\mathbb{C}$ -algebra; so $\mathcal{C}_m$ is equivalent to the category of m -pointed commutative $\mathbb{C}$ -algebras, and initiality is exactly the freeness of $\mathfrak{B}_m$ together with the verification that the distinguished sequence transports along the unique map. The content is modest but the statement is now a theorem with a defined category, defined morphisms, and a proof, rather than an assertion.

8.2 Recognition

Theorem 36. *[Recognition] Let A be a graded connected commutative $\mathbb{C}$ -Hopf algebra with a sequence $\lambda_N \in A_N$, $\lambda_0 = 1$, satisfying:*

- (G). *the λ_N generate A as an algebra;*
- (D). *$\Delta \lambda_N = \sum_{i=0}^N \lambda_i \otimes \lambda_{N-i}$ for all N (divided-power sequence);*
- (T). *$\text{Prim}(A)_j = 0$ for all $j > m$;*
- (I). *$\lambda_1, \dots, \lambda_m$ are algebraically independent over $\mathbb{C}$.*

Then there is a unique isomorphism of graded Hopf algebras $\varphi: \mathfrak{B}_m \rightarrow A$ with $\varphi(\Lambda_N) = \lambda_N$ for all N ; and $\varphi(c_j)$ is the weight- j coefficient of $\log \sum_N \lambda_N t^N$.

Proof. Set $\Lambda^A(t) := \sum_{N \geq 0} \lambda_N t^N \in A[[t]]$. By (D), $\Lambda^A(t)$ is group-like in the completed Hopf algebra $A[[t]]$ (coproduct and counit extended t -adically): $\Delta \Lambda^A = \Lambda^A \otimes \Lambda^A$ read in $(A \otimes A)[[t]]$, $\varepsilon \Lambda^A = 1$. Since $\Lambda^A(t) - 1 \in t A[[t]]$, the logarithm $\ell(t) := \log \Lambda^A(t) = \sum_{j \geq 1} \gamma_j t^j$ is defined, and

$$\Delta \ell = \log((\Lambda^A \otimes 1)(1 \otimes \Lambda^A)) = \log(\Lambda^A \otimes 1) + \log(1 \otimes \Lambda^A) = \ell \otimes 1 + 1 \otimes \ell,$$

the middle equality because the two arguments commute. Hence every γ_j is primitive; also $\gamma_j \in A_j$ because $\lambda_N \in A_N$ and $\log$ preserves the grading of coefficients. By (T), $\gamma_j = 0$ for $j > m$, so

$$\Lambda^A(t) = \exp\left(\sum_{j=1}^m \gamma_j t^j\right), \quad \text{i.e.} \quad \lambda_N = \Lambda_N(\gamma_1, \dots, \gamma_m).$$

Define the algebra map $\varphi: \mathfrak{B}_m \rightarrow A$ by $\varphi(c_j) = \gamma_j$; then $\varphi(\Lambda_N) = \lambda_N$ by the displayed identity (or by Theorem 34, since $(A, \gamma, \lambda) \in \mathcal{C}_m$: apply d/dt to $\Lambda^A = \exp(\sum \gamma_j t^j)$ as in Lemma 2). φ is a morphism of graded Hopf algebras: it preserves grading ($\gamma_j \in A_j$) and sends the primitive generators c_j to primitives, which determines compatibility with Δ , ε , S on all of $\mathfrak{B}_m$ since these are (anti)algebra maps. *Surjectivity*: the image contains every λ_N , which generate by (G). *Injectivity*: $\lambda_1, \dots, \lambda_m$ are polynomials in $\gamma_1, \dots, \gamma_m$ and conversely (the triangularity of Theorem 7 transports along φ), so (I) is equivalent to algebraic independence of $\gamma_1, \dots, \gamma_m$; an algebra map from $\mathbb{C}[c_1, \dots, c_m]$ sending the variables to algebraically independent elements has zero kernel. *Uniqueness*: any graded Hopf map ψ with $\psi(\Lambda_N) = \lambda_N$ satisfies $\psi(\log E(t)) = \ell(t)$ coefficient-wise, i.e. $\psi(c_j) = \gamma_j$, and is therefore φ . $\square$

8.3 Independence of the hypotheses

Proposition 37. *Each of (G), (D), (T), (I) is independent of the remaining three: for each hypothesis there is a triple $(A, (\lambda_N))$ satisfying the other three but not isomorphic to $(\mathfrak{B}_m, (\Lambda_N))$ in the sense of Theorem 36.*

Proof. Fix $m \geq 2$ (for $m = 1$ replace the (I)-example by $A = \mathbb{C}$, $\lambda_N = \delta_{N,0}$, which satisfies (G),(D),(T) but not (I)).

Drop (G). $A = \mathfrak{B}_m[y]$, y primitive of weight 1, with $\lambda_N := \Lambda_N$. (D) holds as before; $\text{Prim}(A)_j = \mathbb{C}c_j$ for $2 \leq j \leq m$, $\text{Prim}(A)_1 = \mathbb{C}c_1 \oplus \mathbb{C}y$, and 0 for $j > m$, so (T) holds; (I) holds. But the λ_N do not generate (y is missing), and $A \not\cong \mathfrak{B}_m$ carrying the sequences (Hilbert series differ).

Drop (I). $A = \mathfrak{B}_m / (c_m) \cong \mathfrak{B}_{m-1}$ with $\lambda_N := \text{image of } \Lambda_N$, i.e. $\Lambda_N(c_1, \dots, c_{m-1}, 0)$. (c_m) is a Hopf ideal (generated by a primitive), so A is a graded connected Hopf algebra and (D) descends; (G) descends; (T) holds a fortiori since $\text{Prim}(A)_j = 0$ for $j \geq m$. But $\lambda_1, \dots, \lambda_m$ satisfy the algebraic relation expressing λ_m through $\lambda_1, \dots, \lambda_{m-1}$ (set $c_m = 0$ in $\Lambda_m = c_m + q_m(c_1, \dots, c_{m-1})$), and $A \not\cong \mathfrak{B}_m$ (transcendence degree $m-1$ vs. m).

Drop (T). $A = \mathfrak{B}_{m+1}$ with its own sequence $\lambda_N := \Lambda_N^{(m+1)}$. (G), (D), (I) hold (Theorems 7, 9; independence of the first m among the first $m+1$). $\text{Prim}(A)_{m+1} = \mathbb{C}c_{m+1} \neq 0$, and no isomorphism onto $\mathfrak{B}_m$ exists (transcendence degree).

Drop (D). $A = \mathfrak{B}_m$ as a graded Hopf algebra (so (T) is unchanged), with the perturbed sequence $\lambda'_N := \Lambda_N$ for $N \neq 2$ and $\lambda'_2 := \Lambda_2 + c_1^2$. (G): $\lambda'_1 = c_1$ and $\lambda'_2 - \frac{3}{2}\lambda_1'^2 = c_2$ recover the generators, and inductively all c_j ; (I): $\lambda'_1, \dots, \lambda'_m$ differ from $\Lambda_1, \dots, \Lambda_m$ by a weight-preserving triangular polynomial change, preserving independence. But

$$\Delta \lambda'_2 - \sum_{i=0}^2 \lambda'_i \otimes \lambda'_{2-i} = \Delta c_1^2 - (c_1^2 \otimes 1 + 1 \otimes c_1^2) = 2 c_1 \otimes c_1 \neq 0,$$

so (D) fails; and no isomorphism φ with $\varphi(\Lambda_N) = \lambda'_N$ can exist, since a Hopf map would transport identity (D), which holds for (Λ_N) and fails for (λ'_N) . $\square$

Remark 38. The axioms now do work in the promised sense: (D)+(T) alone force the generating series into the exponential normal form $\exp(\sum_{j \leq m} \gamma_j t^j)$ (proof of Theorem 36), i.e. they *derive* the universal recurrence rather than postulate it; (G)+(I) then pin the algebra. Grading, derivations (Lemma 5, recovered as ∂_{γ_j}), and the product/coproduct formulas are theorems from the axioms, and the Laurent construction of Section 5 is a model by Theorem 23.

9 Application: rough Heston pricing in the coefficient algebra

We use the notation of the companion papers [CrowleyCGF, CrowleyCW]: the rough Heston characteristic function is $\varphi(a, t) = e^{L(a, t)}$ with $L(a, t) = t G(a, x)$, $x = t^\alpha$, $\alpha = H + \frac{1}{2} \in (0, 1)$; $G(a, \cdot)$ is given by the neighboring-pair series with coefficients $g_j \in \mathbb{C}[a]$ [CrowleyCGF, Thm. 1]; and the diagonal Padé approximant $G_M = A_M / B_M$ ($\deg A_M = \deg B_M = M$) is produced exactly in $\mathbb{C}[a]$ by the polynomial Chebyshev–Wheeler recur-

rence [CrowleyCGF, Thm. 2], [CrowleyCW], with Stahl convergence on the physical ray [CrowleyCGF, Thm. 3]. Define

$$\varphi_M(a, t) := \exp(t G_M(a, t^\alpha)).$$

For fixed a and t , φ_M is the exponential of a rational function of x : exactly the object of this paper.

Theorem 39. *[Local data of φ_M in the coefficient algebra] Fix a and M , and let $\xi_1, \dots, \xi_r$ be the distinct zeros of $B_M(a, \cdot)$ in $\mathbb{C}$, with multiplicities $\mu_1, \dots, \mu_r$, none lying on $[0, x^*]$ where $x^* = t^\alpha$ for the maturity range of interest (guaranteed for a on the pricing contour by the real-ray theorem [CrowleyCGF, Thm. 3] for all M large). Write the exact partial fraction decomposition*

$$t G_M(a, x) = \pi(x) + \sum_{i=1}^r \sum_{j=1}^{\mu_i} \frac{c_j^{(i)}}{(x - \xi_i)^j}, \quad \pi \in \mathbb{C}[x], \deg \pi \leq \deg A_M - \deg B_M + (\text{polynomial part}),$$

all data computable exactly from (A_M, B_M) by division and residue extraction in $\mathbb{C}[a](x)$. Then:

1. $\varphi_M(a, \cdot)$ is analytic on a neighborhood of $[0, x^*]$ in the x -plane and is evaluated there by direct composition $\exp \circ (t A_M / B_M)$; no expansion is needed on the ray and no essential singularity is approached.
2. Around each ξ_i , with $w = x - \xi_i$ and $H_i := t G_M - \sum_j c_j^{(i)} w^{-j}$ analytic on $|w| < r_i$ ($r_i = \min_{i' \neq i} |\xi_{i'} - \xi_i|$),

$$\varphi_M = \left(\sum_{N \geq 0} \Lambda_N(c^{(i)}) w^{-N} \right) \left(\sum_{l \geq 0} b_l^{(i)} w^l \right) \quad (5)$$

$$A_n^{(i)} = \sum_{k \geq \max(0, -n)} \Lambda_k(c^{(i)}) b_{n+k}^{(i)} \quad (6)$$

absolutely convergent Laurent data on $0 < |w| < r_i$, with truncation error controlled by Proposition 25 and with each coefficient sequence P -recursive via (3) applied to $R = t G_M$.

3. *All quantities in (i)–(ii) are algebraic expressions in the Chebyshev–Wheeler output and hence carry rigorous arb enclosures when evaluated in ball arithmetic.*

Proof. (i) is the zero-freeness of B_M on the ray plus continuity of $\exp$. (ii) is Theorem 24 and Proposition 25 applied to the rational function $R(x) = t G_M(a, x)$ at the pole ξ_i (the polynomial part π is absorbed into H_i), together with Theorem 31. (iii) is by inspection: partial fractions, Λ -recurrence (1), and Taylor coefficients of e^{H_i} (themselves obtainable from (3)) are finite exact recurrences. $\square$

9.1 Price convergence without contour deformation

We deliberately do *not* deform any contour across poles of B_M : the poles belong to the approximant, not to φ , and residue exchange across them is not justified. Instead the pricing contour is held fixed inside the strip of analyticity of the exact characteristic function, and convergence is proved by domination.

Definition 40. [Lewis-type pricing functional] Fix maturity $T > 0$, $x^* = T^\alpha$, log-strike k , and a horizontal contour $\Im a = \eta_0$ interior to the moment-analyticity strip of $\varphi(\cdot, T)$. A Lewis-type functional is

$$P[\psi] = c_0 + \int_{\mathbb{R}} K(u) \psi(u + i\eta_0) du, \quad |K(u)| \leq \frac{C_K}{1 + u^2},$$

with c_0, K depending only on $(S_0, K, r, T, k, \eta_0)$; the call price is $P[\varphi(\cdot, T)]$ for the standard choices of (c_0, K, η_0) (Lewis [Lewis2001]; Carr–Madan [CarrMadan]; Gil-Pelaez [GilPelaez]).

Theorem 41. [Convergence of Padé prices] Assume:

1. for every $u \in \mathbb{R}$, $B_M(u + i\eta_0, \cdot)$ is zero-free on $[0, x^*]$ for all $M \geq M_0(u)$, and $\varphi_M(u + i\eta_0, T) \rightarrow \varphi(u + i\eta_0, T)$ as $M \rightarrow \infty$;
2. there exists $\Phi \in L^1\left(\mathbb{R}, \frac{du}{1+u^2}\right)$ and M_1 with $|\varphi_M(u + i\eta_0, T)| \leq \Phi(u)$ for all $M \geq M_1$ and a.e. u .

Then $P[\varphi_M(\cdot, T)] \rightarrow P[\varphi(\cdot, T)]$: Padé prices converge to exact prices. Moreover (H1) holds pointwise on the contour as a consequence of the real-ray Stahl theorem [CrowleyCGF, Thm. 3] applied at each fixed $a = u + i\eta_0$ (convergence of $G_M(a, \cdot) \rightarrow G(a, \cdot)$ on a neighborhood of $[0, x^*]$, hence of the exponentials), so the only unproved input is the domination (H2).

Proof. For $M \geq \max(M_0(u), M_1)$ the integrand $K(u) \varphi_M(u + i\eta_0, T)$ converges pointwise to $K(u) \varphi(u + i\eta_0, T)$ by (H1) and continuity of nothing further (the integrand is the object itself), and is dominated by $C_K \Phi(u)/(1+u^2) \in L^1$ by (H2). Apply dominated convergence. The claim about (H1): for fixed a on the contour, Stahl convergence in capacity together with zero-freeness of B_M on a fixed neighborhood $\Omega \supset [0, x^*]$ upgrades, by the standard capacity-to-uniform argument on compacts avoiding the poles (Stahl [Stahl, Thm. 1 and §6]), to uniform convergence $G_M \rightarrow G$ on $[0, x^*]$; then $\varphi_M = \exp(TG_M(a, T^\alpha)) \rightarrow \exp(TG(a, T^\alpha)) = \varphi$. $\square$

Remark 42. [Status of (H2)] (H2) is the precise statement of what remains open on the analytic side. It is implied by a uniform-in- M , integrable-in- u bound on $\Re T G_M(u + i\eta_0, T^\alpha)$ from above; numerically, the implemented pipeline exhibits $|\varphi_M|$ dominated by the exact $|\varphi|$ -envelope (Gaussian-type decay in u from the leading cumulant $g_0 = v_0 F_0$ [CrowleyCGF, Cor. 1.1]) uniformly at machine-verified precision, consistently with the end-to-end validation of the pricing series to 1.1×10^{-14} at $M = 20$ (40 coefficients) [CrowleyCGF]. A proof of (H2) is Open Problem 1; Theorem 41 isolates it as the exact remaining gap, replacing the invalid contour-deformation route.

Proposition 43. *[Cumulants in the algebra] Every $c_k(a)$ and hence every $g_j(a)$ is a polynomial in a ([CrowleyCW, Lem. 2.1–2.2]), and for $0 \leq t < t_0$ with t_0^α less than the shared radius [CrowleyCGF, Cor. 1.2] uniformly for a in a neighborhood of 0 , the cumulants of $\log S_T$ are the absolutely convergent series*

$$\kappa_n(t) = \partial_a^n L(a, t)|_{a=0} = t \sum_{j \geq 0} (\partial_a^n g_j)(0) t^{\alpha j},$$

with each $(\partial_a^n g_j)(0)$ a finite exact expression obtained by differentiating the Gamma-ratio recurrence in a . Consequently the cumulant data feeding any transform- or expansion-based pricing representation lies in $\mathbb{C}$ -linear combinations of the same neighboring-pair objects, computed without approximation.

Proof. Polynomiality in a is the coefficient-ring statement of [CrowleyCW]. For $|a| \leq \epsilon$ and $t \leq t_1 < t_0$ the series $\sum g_j(a) t^{\alpha j}$ converges uniformly (Cauchy majorization from the shared radius, which is locally uniform in a by the analyticity argument of [CrowleyCGF, Cor. 1.2]); a locally uniformly convergent series of polynomials in a may be differentiated termwise in a any number of times (Weierstrass), giving the display. Each $\partial_a^n g_j(0)$ is a finite combination of $\partial_a^n c_j(0), \partial_a^n c_{j+1}(0)$ via [CrowleyCGF, Eq. (2)], and $\partial_a^n c_k(0)$ obeys the recurrence obtained by Leibniz differentiation of the Gamma-ratio recurrence, closed because F_0, β are polynomials in a . $\square$

Algorithm 44. *[Coefficient-algebra evaluation layer for arb4j] Input: $(A_M, B_M) \in \mathbb{C}[a][x]^2$ from the Chebyshev–Wheeler pipeline; contour height η_0 ; maturity T ; working precision; truncation bandwidth K .*

1. *For each quadrature node $a = u + i\eta_0$: evaluate $B_M(a, x)$ on $[0, T^\alpha]$ in ball arithmetic; certify zero-freeness (interval Descartes/argument principle). Cost $O(M)$ per node.*
2. *Price by fixed-contour quadrature of $K(u) \exp(T A_M / B_M)$ (Theorem 41); rigorous tails from the kernel bound. Cost: nodes $\times O(M)$.*
3. *(Diagnostics / analytic continuation off the ray.) For each required pole ξ_i : partial fractions to get $c^{(i)}$ ($O(M^2)$ exact ops); Λ -sequence by (1) ($O(mK)$ ops); Taylor data $b_l^{(i)}$ of e^{H_i} by (3) ($O((d_U + d_V)K)$ ops); assemble $A_n^{(i)}$ with certified tail via Proposition 25.*
4. *Cumulant channel: $\partial_a^n c_k(0)$ by the differentiated Gamma-ratio recurrence ($O(nM)$ per order), then κ_n by Proposition 43.*

All steps are exact recurrences or ball-arithmetic evaluations with explicit error terms; no Hankel solve, no numerically unstable step, matching the stability profile of [CrowleyCW, Thm. 10.2].

10 Open problems

Problem 1. [Domination] Prove hypothesis (H2) of Theorem 41: a uniform-in- M integrable majorant for $|\varphi_M|$ on the Lewis contour, e.g. via a uniform-in- a version of the Stahl rate on the contour combined

with the exact Gaussian-type decay of φ .

Problem 2. [Resurgent structure] e^Π is the exponential factor of the Hukuhara–Levelt–Turrittin normal form of the rank-one connection $d - dR$ at the irregular singularity a ([vdPS, Ch. 3]); this much is classical. Construct the action of the alien derivations of Écalle on the $\mathfrak{B}_m$ -module of Theorem 24 — the natural conjecture, suggested by Lemma 5, is that singular directions act through the lowering operators ∂_{c_j} — and determine the Stokes data of φ_M in terms of the $c^{(i)}$ (framework: Mitschi–Sauzin [MitschiSauzin]; Costin [Costin]). No such statement is asserted here as a theorem.

Problem 3. [Exponential periods] $e^R dz$ is an exponential-period integrand; the category of exponential motives (Fresán–Jossen [FresanJossen]) attaches cohomological invariants to $(\mathbb{A}^1, R)$. Position $\mathfrak{B}_m$ relative to the de Rham realization of the rank-one exponential connection: is the coefficient module of Theorem 24 a concrete model of a known object there?

Problem 4. [q -lift] Lift (Λ_N) along the Hall-algebra isomorphism of Remark 13 to the Hall–Littlewood level: identify the q -analogue of the universal recurrence and its analytic meaning, if any, for q -deformed exponentials of principal parts.

Problem 5. [Non-reduction] Prove (or refute) that for generic (c_1, c_2, c_3) the sequence $(\Lambda_N(c_1, c_2, c_3))_N$ is not expressible as $(\rho^N F(\sigma N + \tau))_N$ for any single classical hypergeometric-type function F of one variable — i.e. make Remark 30 a theorem, plausibly via differential Galois groups of the order-3 holonomic equation from (1).

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